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Onderzoek de convergentie van de reeks $\sum_{n=0}^{\infty} \frac{(-1)^n 7^n}{n!}$

d'Alembert

Stel $x_n = \frac{(-1)^n 7^n}{n!}$, dan is $|x_n| = \frac{7^n}{n!}$

Stel $|x_{n+1}| = \frac{7^{n+1}}{(n+1)!}$

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \left| \frac{x_{n+1}}{x_n} \right| \\ &= \lim_{n \rightarrow \infty} \left(\frac{7^{n+1}}{(n+1)!} \cdot \frac{n!}{7^n} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{7 \cdot 7^n}{(n+1) \cdot n!} \cdot \frac{n!}{7^n} \right) \\ &= \lim_{n \rightarrow \infty} \frac{7}{n+1} \\ &= 0 \end{aligned}$$

Aangezien $L = 0 < 1$, is de reeks absoluut convergent.

References